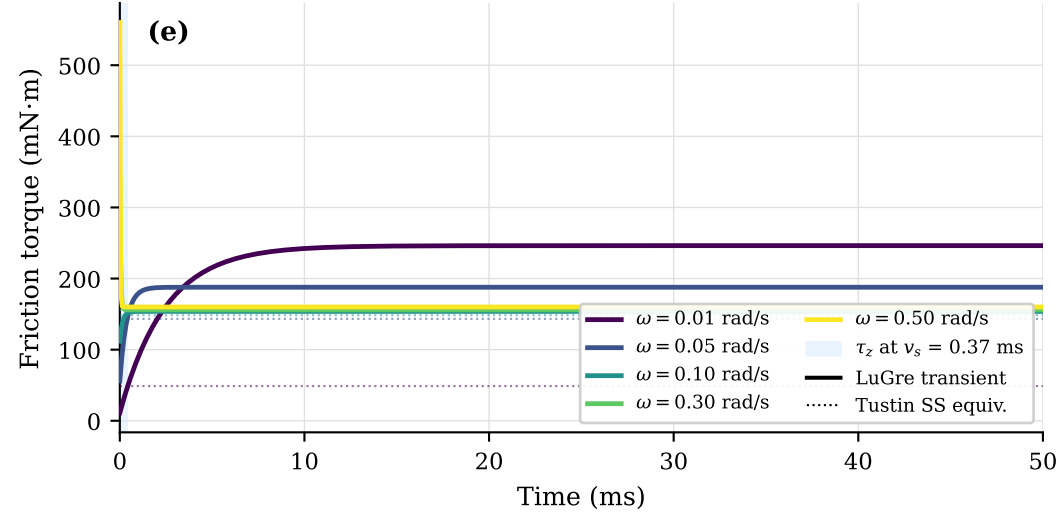
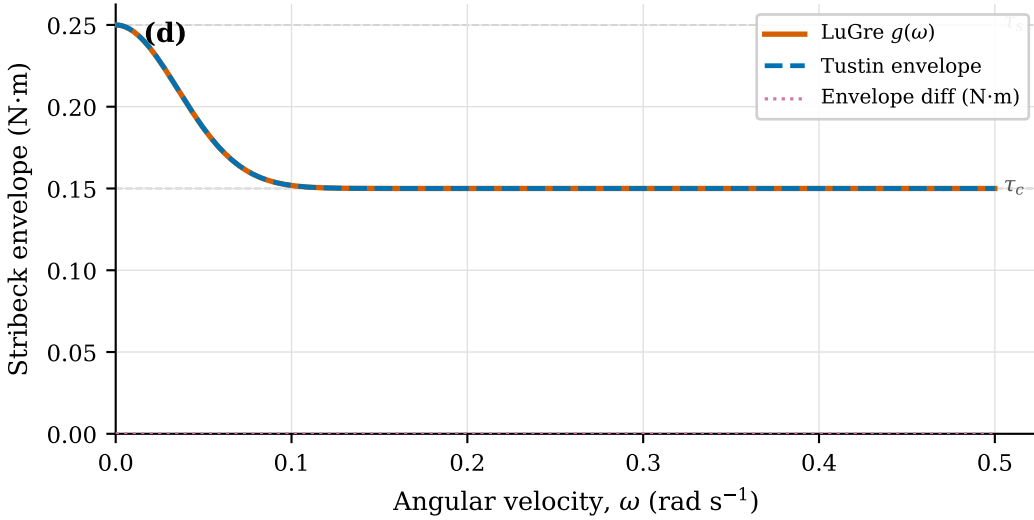
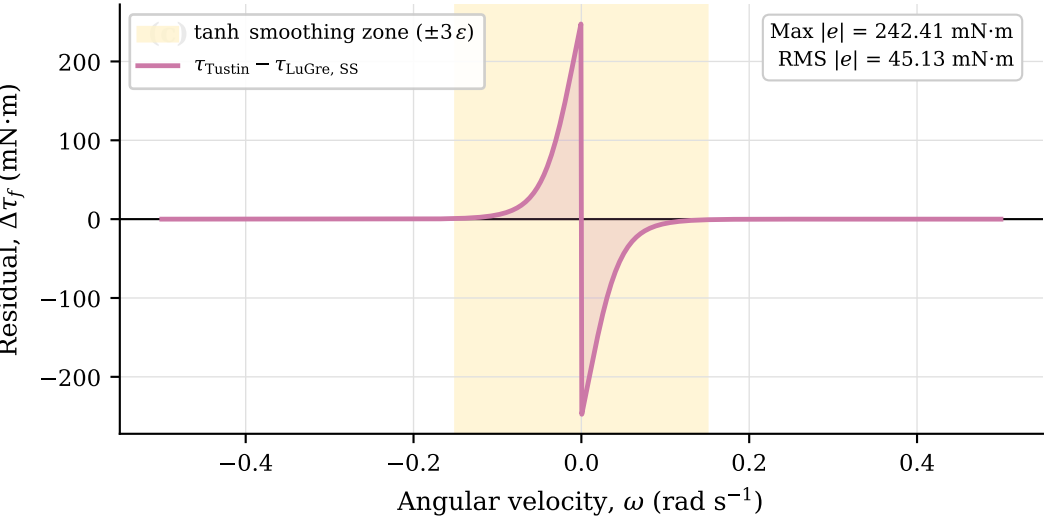
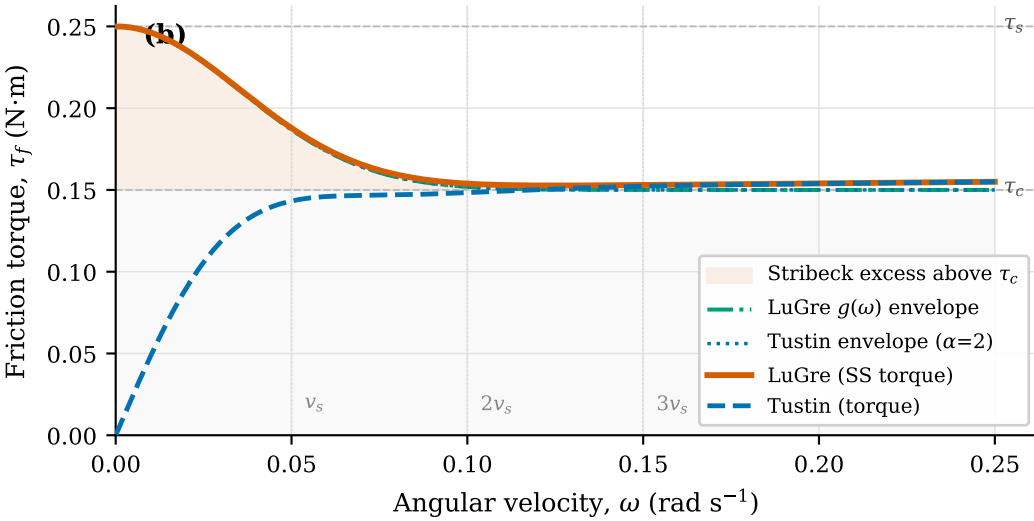
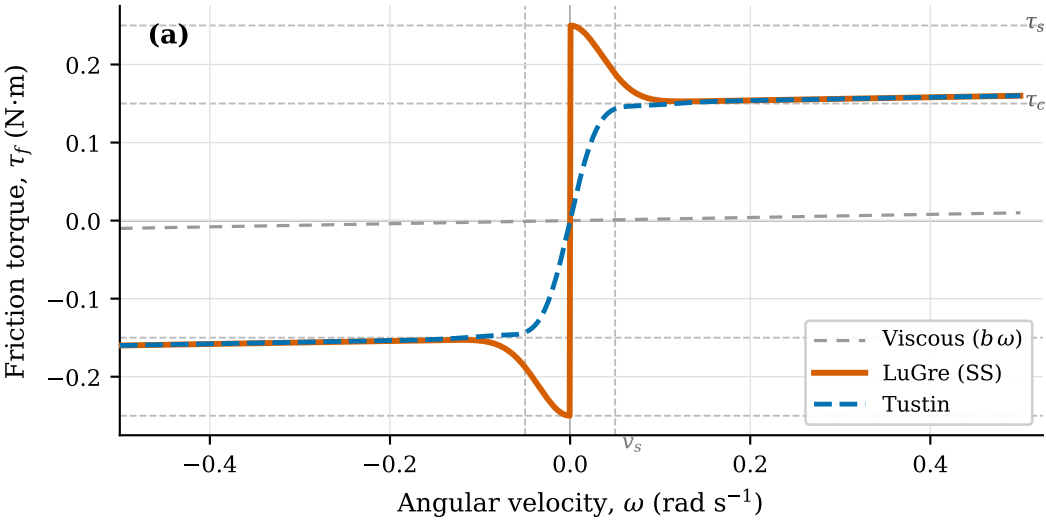


Friction Model Comparison — Azimuth Gimbal | Steady-state match: confirmed



(f)

Parameter	Tustin	LuGre	Match
τ_c (N·m)	0.15	0.15	Yes
τ_s (N·m)	0.25	0.25	Yes
v_s (rad s ⁻¹)	0.05	0.05	Yes
b/σ_2 (N·m s)	0.02	0.02	Yes
α (Tustin only)	2	2 (fixed)	Yes
σ_0 (N·m rad ⁻¹)	—	1.000e+04	transient
σ_1 (N·m s)	—	1	transient
ε (rad s ⁻¹)	0.05	—	smoothing
$f_{\text{bristle}} = \frac{1}{\varepsilon} \sqrt{\frac{\sigma_0}{\pi}}$ (Hz)	—	15.9	dynamics
$f_{\text{NDOB}} = \lambda/(2\pi)$ (Hz)	—	—	14.3

Steady-state match metrics

Max $|\Delta\tau| = 242.407$ mN·m (3218.12%)
RMS $|\Delta\tau| = 45.130$ mN·m (197.54%)

Bristle dynamics at v_s

$\tau_z = g(v_s)/(\sigma_0 v_s) = 0.374$ ms
 $\omega_n = \sqrt{\sigma_0} = 100$ rad s⁻¹